

Algebra II with Trigonometry

Chapter 7.3

Olympiad Problems (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4592201_alg-2trig-h-chp-7-3-note-docx.pdf](#)

Problems

1. Evaluate exactly:

$$\cos 15^\circ \cos 45^\circ \cos 75^\circ.$$

2. Find the exact value of

$$\frac{\sin 10^\circ + \sin 50^\circ + \sin 70^\circ}{\cos 10^\circ + \cos 50^\circ + \cos 70^\circ}.$$

3. Suppose $0 < \theta < \frac{\pi}{2}$ and

$$\sin \theta + \cos \theta = \frac{5}{4}.$$

Find the exact value of $\tan 2\theta$.

4. Evaluate exactly:

$$\sqrt{\frac{1 - \cos 20^\circ}{1 + \cos 20^\circ}} + \sqrt{\frac{1 - \cos 40^\circ}{1 + \cos 40^\circ}} + \sqrt{\frac{1 - \cos 80^\circ}{1 + \cos 80^\circ}}.$$

5. Find the exact value of

$$\sin 20^\circ \sin 40^\circ \sin 80^\circ.$$

6. Let

$$S = \cos x + \cos 3x + \cos 5x + \cos 7x.$$

If $x = \frac{\pi}{8}$, compute S exactly.